Lab 6: bench checklist

Bulb and temperatures

Record the **bulb number** and use the same bulb for all runs. Read room temperature and the glass temperature at the marked equator point (emissivity 0.95 if adjustable). The reading is the **glass, not the vanes**. A **cooled start** is glass within 1 °C of room temperature with the rotor at rest. Once, find the largest gap at which the rotor starts within 30 s.

Gaps and light

Use the course LED lamp at fixed power. Set the gap from its light-emitting front surface to the nearest glass surface at rotor height with spacer blocks or a ruler: **1.0 cm, 1.5 cm, 2.0 cm, then repeat 1.0 cm after cooling**. If the start gap is below 2.0 cm, use it as the third gap and log the actual value. Measure lux with the sensor at the rotor-centre position, bulb lifted off its mark, facing the lamp. Keep the background **25 cm behind the rear glass surface** and out of the strongest beam. Measure the lamp-face temperature once.

Camera

1280 × 1024 pixels; 1000 capture frames/s. Other settings are preset. Check a short movie, verify p over one full turn, and confirm that the recording holds at least seven passage marks plus 10 s after light-off.

Record

Glass temperature, then start recording before light-on. After seven or more passages, **switch the lamp off and keep recording**. Glass temperature again. Save 1cm_1.mp4, 1p5cm.mp4, 2cm.mp4 and 1cm_repeat.mp4 as H.264 with Text Overlay off. Note light-on and light-off frames and whether the rotor reversed. Wait for a cooled start before the next run.

Analyze

Mark_Lab6_Passage_Frames.m saves passage CSVs; run Run_Lab6_Radiometer.m for each. For 1cm_1, also mark the post-light-off passages as 1cm_1_coast_passages.csv and fit ln(rpm) versus time for T. Complete MIE402_Lab6_Run_Log.csv and the prefilled Lab6_Comparison_Log.csv, then run Compare_Lab6_Illumination_Runs.m with the same light-on window for all rows.

Keep

Raw movies, passage CSVs including the coast-down file, both logs and exported figures. The main guide lists the report figures and the required Q6 connection.